Victor Kneider

Software Engineer · Educator · Researcher
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Profile

Software engineer focused on professional software design and architecture. I design, build, teach, and share professional software through production-oriented projects, university teaching, and open-source work.

Professional Experience

Computer Engineering Professor, Universidad Rafael Urdaneta (URU) – Maracaibo, Venezuela 2024 - Present

University professor in computer engineering programs, teaching software development, web programming, and component-based architecture.

- Teach Programming 2, Client-Side Web Development, and Component-Based Programming.
- Develop tools that support the School's growth, automation, and process optimization.
- Automated make-up class scheduling by mapping the schedules of 30+ students in a section, replacing manual review and enabling professors to identify the best recovery times—functionality not provided by the institution's existing system.
- Developed a shared information platform designed to serve 200+ URU professors, requested by the Graduate Department and delivered to the School.
- Advised 3 undergraduate theses; all 3 were recommended for publication distinction, while mentoring 120+ students through academic projects.
- Research and develop the **Slice.js** web framework.

Computer Engineer & Framework Creator, Freelance / Personal Projects – Remote 2020 - Present

Independent software engineering work focused on web architecture, open source, educational tools, and the **Slice.js** framework.

- Design and develop the architecture of **Slice.js**, a component-based web framework.
- Create responsive web applications using modern JavaScript and **Web Components**.
- Develop educational tools and browser-based games for academic use.
- Maintain open-source repositories and technical documentation.

Computer Engineer — AIT Department Intern, Polipropileno de Venezuela (PROPILVEN S.A.) – Venezuela 2023 · 6 mo.

Internship in the Information and Technology department, contributing to automation systems and software solutions for industrial processes.

- Developed and maintained automation software systems.
- Assisted with IT infrastructure management and troubleshooting.
- Created documentation for technical processes and procedures.
- Participated in system analysis and optimization projects.
- Successfully completed the six-month internship program.

Education

Universidad Rafael Urdaneta (URU), Bachelor's degree in Computer Engineering – Maracaibo, Venezuela 2020 - 2024

Computer Engineering degree focused on software development, algorithms, data structures, databases, and system design.

- Graduated Magna Cum Laude with academic honors.
- Created **Slice.js** as part of academic and research work.
- Recognized for perfect attendance by the School of Computer Engineering at Universidad Rafael Urdaneta.

Publications

Desarrollo de un marco de trabajo basado en componentes para la construcción de clientes web: Slice.js July 2025

Research article presenting the design and development of a component-based framework for building web clients, with an emphasis on intuitive architecture, decoupled components, maintainability, and backward compatibility.
Julio A. Graterol Bracho, *Victor J. Kneider Al Nahi*, Jubert J. Pérez Zabala
10.5281/zenodo.17651991 (Revista tecnocientífica URU, no. 28, pp. 86–98)

Selected Projects

Slice.js Framework Jan 2024

Component-based web framework designed to explore modern frontend architecture without virtual DOM overhead.

- Includes reactive state, scoped styling, dependency injection, and a custom CLI for scaffolding and bundling.
- Built with JavaScript, **Web Components**, HTML5, and CSS3.
- Published as open source at slicejs.com.

Parish Automated Ecosystem June 2024

Zero-cost digital ecosystem for San Ramón Nonato Parish, powered by an automated content pipeline.

- Uses Google Sheets and Forms as a headless CMS for administrative data.
- Transforms source data through custom ETL scripts into a performant web experience.
- Built around process automation, cost optimization, and Vercel deployment.

Conclave July 2026

Client-side group decision-making tool for assignment, voting, ranking, and free-text brainstorming.

- Lets teams compare independent responses and reach consensus without endless meetings.
- Keeps all decision data in the browser and supports reusable decision templates.
- Built with **Slice.js**, JavaScript, **Web Components**, Express, and localStorage.

Technical Skills

Frontend: TypeScript, HTML5, CSS3, JavaScript, React, Next.js, **Slice.js**, **Web Components**, Tailwind CSS, Chakra UI

Backend & Architecture: Software architecture, event-driven systems, API design, **Node.js**, Express, **Python**, FastAPI, Celery Workers, REST APIs, gRPC, tRPC, Uvicorn, Gunicorn, Cloud Workers

Cloud, Data & Deployment: Azure, Vercel, Railway, **PostgreSQL**, MongoDB, Docker, Redis, Linux, SSH, CI/CD

AI & Modern Development: AI Early Adopter, MCP Builder, Model Context Protocol, Claude Code, Cursor, generative AI, prompt engineering, Git, Makefiles

Testing & Quality: Vitest, Playwright, unit testing, integration testing, component testing, End-to-End (E2E) testing, Storybook

Languages

Spanish: Native

English: Advanced (C1) — Centro Electrónico de Idiomas, grade 92/100

Teaching

Programming 2 – Rafael Urdaneta University (URU)

Structured and object-oriented programming in C++ with modularization, file handling, data structures, and separation of logic from presentation.

- Functions, custom libraries, structures, text and binary files.
- Classes, encapsulation, object persistence, pointers, linked lists, stacks, and queues.

Client-Side Web Development – Rafael Urdaneta University (URU)

Web development with HTML, CSS, and JavaScript without frameworks, from web fundamentals to API integration and browser persistence.

- Semantic HTML, forms, responsive layouts, DOM manipulation, events, and client-side validation.
- Promises, async/await, REST APIs, Custom Elements, SPA navigation, LocalStorage, IndexedDB, and Chart.js.

Component-Based Programming – Rafael Urdaneta University (URU)

Advanced elective on component-based frontend architecture, communication patterns, design systems, and design patterns.

- **Web Components**, component lifecycles, Props, Context, Event Bus, routers, themes, and global managers.
- Atomic Design, Compound Components, Headless Components, Strategy, Factory, Observer, Repository, and Middleware patterns.
- Apply unit, integration, and end-to-end testing to components, with Storybook for development and documentation.

Certifications & Credentials

Specialist in Higher Education Didactics Aug 2025

World Association for Curriculum Quality (WACQ)

- Higher education didactics, curriculum design, learning assessment, and instructional planning.
- Level 4 credential aligned with the European Qualifications Framework.

Advanced Artificial Intelligence II Oct 2024

Universidad Rafael Urdaneta

- Advanced artificial intelligence and applied AI in computer engineering.

Fullstack Developer Oct 2022

Social Oplesk

- Frontend, backend, APIs, JavaScript, **Node.js**, and database design.

ChatBOT Hackathon — Second Place July 2022

Social Oplesk — team project

- Built a full-cloud serverless chatbot solution with **Node.js**.
- Deployed across Railway and Vercel as a collaborative team project.

Catholic Leadership Workshop July 2022

Acción Católica de Venezuela

• Leadership, community service, teamwork, and ethics.